

POLYTECHNICS MAURITIUS

BUILDING INTERNET OF THINGS (DET 2105C)

SMART 4-WAY IoT ADAPTIVE TRAFFIC CONTROL SYSTEM



PROJECT REPORT

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Project	IoT Based Traffic Control
Academic Year	2026

Wokwi Simulation: <https://wokwi.com/projects/471082310152702977>

Executive Summary

This project presents a Smart 4-Way IoT Adaptive Traffic Control System developed using an ESP32 microcontroller. The system improves traffic-flow efficiency by replacing a purely fixed-time traffic sequence with an adaptive approach based on real-time vehicle detection. Four HC-SR04 ultrasonic sensors monitor the North, East, South and West roads, while twelve LEDs represent the red, yellow and green traffic signals.

The main feature is distance-based priority. When only one road has vehicles, that road can remain green for longer instead of wasting time on empty roads. When several roads contain vehicles, the road with the nearest detected vehicle receives higher priority. Safety is maintained through a fixed yellow phase and an all-red transition period. A pedestrian push button can request a dedicated all-red crossing period accompanied by a buzzer.

The system was tested in Wokwi and then transferred to a physical maquette. The simulation successfully demonstrated the intended adaptive traffic behaviour and provided a practical basis for the working physical prototype.

1. Introduction

Traffic congestion can be made worse when traffic lights operate using fixed timing regardless of the actual number of vehicles waiting on each road. A road with no vehicles may receive the same green time as a heavily occupied road. This project addresses the problem by using ultrasonic sensors to detect vehicle presence and distance, allowing the ESP32 to make adaptive traffic decisions.

The project demonstrates how IoT sensing, embedded programming and automated decision-making can be combined to create a responsive four-way traffic-control prototype.

2. Aim and Objectives

The main aim is to design and implement an adaptive four-way traffic-control prototype that responds to real-time vehicle presence and distance.

- Use an ESP32 as the central controller.
- Detect vehicles on four roads using four HC-SR04 ultrasonic sensors.
- Control twelve LEDs representing four traffic-light groups.
- Give priority to the road with the nearest detected vehicle.

- Allow a single occupied road to remain green for longer.
- Reduce green time on empty roads.
- Provide yellow and all-red safety phases.
- Provide a pedestrian request using a push button and buzzer.
- Validate the design through Wokwi simulation before physical implementation.

3. System Overview

The system consists of four traffic-light groups, one for each road. Each group contains red, yellow and green LEDs. An HC-SR04 ultrasonic sensor is assigned to each road. The sensors share one trigger output from the ESP32, while each sensor has an individual echo input.

A vehicle is considered present when its measured distance is below 50 cm. If several roads are occupied, the smallest distance is treated as the highest priority. This creates a simple distance-priority mechanism suitable for a physical traffic maquette.

4. Hardware Components

Component	Purpose
ESP32 Microcontroller Board	Main processing and control unit.
4 × HC-SR04 Ultrasonic Sensors	Detect vehicle presence and measure distance.
12 × Traffic LEDs	Represent four red/yellow/green traffic-light groups.
12 × 220 Ω Resistors	Limit LED current.
Push Button	Requests pedestrian crossing.
TMB12A05 Active Buzzer	Provides audible alerts.
Breadboard and Jumper Wires	Prototype interconnections.
10 kΩ Resistor	External pull-up for GPIO34 button input.
1 kΩ + 2 kΩ Resistors	Voltage dividers for HC-SR04 echo signals.

5. Final ESP32 Pin Map

Road	Red	Yellow	Green	Echo
North	GPIO 2	GPIO 4	GPIO 5	GPIO 32
East	GPIO 12	GPIO 13	GPIO 14	GPIO 33
South	GPIO 15	GPIO 16	GPIO 17	GPIO 35
West	GPIO 18	GPIO 19	GPIO 25	GPIO 39 (VN)
Shared HC-SR04 Trigger	—	—	—	GPIO 27
Active Buzzer	—	—	—	GPIO 26

Pedestrian Button	—	—	—	GPIO 34
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GPIO34 is input-only and has no internal pull-up, so the physical button uses an external 10 k Ω pull-up resistor. The HC-SR04 echo outputs are reduced using resistor voltage dividers before reaching the ESP32 inputs.

6. Circuit Operation

1. The four HC-SR04 sensors measure the distance of objects in front of each road.
2. A distance below 50 cm is interpreted as vehicle presence.
3. If no road has a vehicle, the controller follows its normal sequence.
4. If only one road has traffic, that road receives priority and can remain green for an extended period.
5. If multiple roads have traffic, the road with the smallest measured distance receives priority.
6. An empty road receives a short three-second green phase, while an occupied road can remain green for longer.
7. After green, the selected road changes to yellow for two seconds.
8. All traffic lights then become red for a one-second safety gap.
9. If the pedestrian button is pressed, all roads remain red for five seconds while the buzzer pulses.
10. The sensors are checked again and the next priority road is selected dynamically.

7. Adaptive Distance-Priority Algorithm

The most important feature is the adaptive priority mechanism. Instead of treating all four roads equally, the controller uses ultrasonic distance measurements to estimate which approach requires attention.

Traffic condition	Decision	Green behaviour
No vehicles	Continue sequence	Normal/short phase
One road occupied	Give that road priority	Can extend up to 15 seconds
Several roads occupied	Select smallest distance	Closest vehicle gets priority
Current road becomes empty	Re-evaluate traffic	Priority can move safely
Pedestrian request	All roads red	Five-second crossing

The prototype uses a 50 cm detection threshold, a minimum occupied-road green duration of five seconds and a maximum green duration of fifteen seconds. An empty road uses three seconds. Yellow is fixed at two seconds and the normal all-red safety gap is one second.

8. Wokwi Simulation

The complete circuit was simulated and tested using Wokwi before the physical maquette was assembled. The simulation allowed the traffic lights, ultrasonic sensors, button and buzzer behaviour to be checked in a controlled environment. It was also used to validate the adaptive priority concept before transferring the design to physical hardware.

Simulation link: <https://wokwi.com/projects/471082310152702977>

9. Testing and Results

Test	Method	Observed result	Status
Single-road traffic	Vehicle on one road; others empty.	Occupied road receives priority and can stay green longer.	Successful
Multiple-road traffic	Vehicles on two or more roads.	Nearest detected vehicle receives priority.	Successful
Empty road	No vehicle detected.	Short green phase prevents unnecessary delay.	Successful
Vehicle departure	Remove vehicle from active road.	Controller re-evaluates traffic.	Successful
Pedestrian request	Press push button.	All roads red and buzzer signals crossing.	Successful
Safety transition	End green phase.	Yellow followed by all-red gap.	Successful

10. Safety and Engineering Considerations

The controller prevents conflicting green lights by switching all roads to red before assigning a new green phase. A two-second yellow phase provides a transition between green and red, followed by a one-second all-red gap. A pedestrian request also forces an all-red state before the crossing period begins.

For the physical prototype, the HC-SR04 echo signals are reduced using resistor voltage dividers before reaching the ESP32. Traffic LEDs use current-limiting resistors, and GPIO34 uses an external 10 kΩ pull-up resistor.

11. Limitations

- HC-SR04 sensors detect objects by distance and cannot identify real vehicles without additional sensing.
- The shared trigger arrangement requires sequential sensor readings to reduce ultrasonic interference.
- The 50 cm threshold is intended for the prototype and would require calibration for real roads.
- The final MVP does not include a mobile application or cloud service.
- The maquette represents a simplified traffic environment and does not model every real-world condition.

12. Future Enhancements

- Add an IoT dashboard or mobile application for monitoring and configuration.
- Add a display showing traffic density, distance and signal state.
- Introduce traffic-density scoring rather than distance alone.
- Add emergency-vehicle detection and emergency priority.
- Store traffic statistics for later analysis.
- Use wireless connectivity for alerts and performance data.
- Integrate cameras or additional sensors for vehicle classification.

13. Conclusion

The Smart 4-Way IoT Adaptive Traffic Control System successfully demonstrates how an ESP32 and ultrasonic sensors can create an intelligent traffic-control prototype. The project moves beyond a fixed-time traffic light by considering real-time vehicle presence and distance when selecting the next priority road.

The main benefit is the ability to avoid unnecessary waiting on empty roads. When only one road has traffic, it can receive extended green time. When several roads are occupied, the closest vehicle is prioritised. Together with the yellow transition, all-red safety gap and pedestrian crossing function, these features provide a practical MVP for adaptive IoT traffic management.

The successful Wokwi simulation provided an effective development and testing environment before physical implementation. The project therefore demonstrates the progression from IoT concept and software logic to simulation and physical maquette.

14. References

- Wokwi. (2026). Smart 4-Way IoT Adaptive Traffic Control System simulation.
<https://wokwi.com/projects/471082310152702977>
- Espressif Systems. ESP32 technical documentation.
- HC-SR04 ultrasonic distance sensor technical documentation.

Appendix A – Final MVP Configuration

- Controller: ESP32 DevKit.
- Roads: North, East, South and West.
- Traffic outputs: 12 LEDs (4 red, 4 yellow, 4 green).
- Vehicle detection: 4 HC-SR04 ultrasonic sensors.
- Shared trigger: GPIO27.
- Echo: GPIO32, GPIO33, GPIO35 and GPIO39/VN.
- Buzzer: GPIO26.
- Pedestrian button: GPIO34 with external 10 k Ω pull-up.
- Detection threshold: 50 cm.
- Occupied-road green: 5–15 seconds.
- Empty-road green: 3 seconds.
- Yellow: 2 seconds.
- Normal all-red gap: 1 second.
- Pedestrian crossing: 5 seconds.
- Final MVP excludes LCD, potentiometer and Blynk.